WEATHERING, MASS MOVEMENT, GROUNDWATER / INDIA EROSION-LANDSLIDES-GROUNDWATER

EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM → WEATHERING TAXONOMY, CONTROLS, REGOLITH → SOIL EROSION SEQUENCE AND → MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY → INDIA LANDSLIDE GEOGRAPHY, RISK → GROUNDWATER ARCHITECTURE FROM RECHARGE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ○ SUBORDINATE EXTRA ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g10 and all prior artifacts unchanged

00

STAGE 00

EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM

EXOGENIC SOURCE-TRANSFER-STORAGE SYSTEM · EXOGENIC LANDSCAPE SYSTEM · HOW IS ROCK CONVERTED INTO LANDFORM? · ROCK EXPOSED AT SURFACE WEATHERING IN PLACE REGOLITH AND · DENUDATION IS COLLECTIVE LOWERING; WEATHERING, EROSION AND MASS WASTING · MUST REMEMBER · WEATHERING PREPARES MATERIAL IN SITU, MASS MOVEMENT TRANSFERS IT

EXOGENIC LANDSCAPE SYSTEM → CENTRAL QUESTION: How is rock converted into landform? → ROCK EXPOSED AT SURFACE weathering in place regolith and loose material supplied gravity driven transfer agent driven → transfer mass wasting downslope water, wind, ice, waves erosion + transport deposition temporary storage remobilisation → Denudation is collective lowering: weathering, erosion and mass wasting are linked but distinct processes.

CORE CLAIM

- EXOGENIC LANDSCAPE SYSTEM
- CENTRAL QUESTION: How is rock converted into landform?
- ROCK EXPOSED AT SURFACE weathering in place regolith and loose material supplied gravity driven transfer agent driven

MECHANISM

- transfer mass wasting downslope water, wind, ice, waves erosion + transport deposition temporary storage remobilisation
- Denudation is collective lowering: weathering, erosion and mass wasting are linked but distinct processes.
- MUST REMEMBER: Weathering prepares material in situ, mass movement transfers it downslope, running water removes it, and groundwater

CONSEQUENCE / LIMIT

- Indian analysis links lithology, slope, rainfall, land use and pumping.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Denudation is a connected source-transfer-storage system, but weathering, erosion and mass wasting must still be distinguished by whether material remains in place, is carried by an agent, or moves mainly under gravity.

01

STAGE 01

WEATHERING TAXONOMY, CONTROLS, REGOLITH AND SOIL FORMATION

WEATHERING TAXONOMY, CONTROLS, REGOLITH AND SOIL FORMATION · MECHANICAL CHEMICAL BIOLOGICAL THERMAL SOLUTION ROOTS/BURROWS UNLOADING CARBONATION ORGANIC · REGOLITH ORGANIC INPUTS + ORGANISMS + TRANSLOCATION + TIME · R REMOVAL GAIN PARENT ROCK · CONTROLS CONVERGE · MINERALOGY + JOINTS/BEDDING + CLIMATE + RELIEF DRAINAGE + · BLACK SOIL LINK · DECCAN BASALT PARENT, SHRINK-SWELL CLAY, DEEP DRY CRACKS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation REGOLITH organic inputs + organisms + translocation + time	O → A/E → B → C → R removal gain parent rock CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief drainage + organisms + time + land use	Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks parent material does not alone determine every soil property.

CORE CLAIM

- MECHANICAL CHEMICAL BIOLOGICAL thermal solution roots/burrows unloading carbonation organic acids frost hydration human exposure salt hydrolysis wet-dry oxidation
- REGOLITH organic inputs + organisms + translocation + time

MECHANISM

- O → A/E → B → C → R removal gain parent rock
- CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief drainage + organisms + time + land use

CONSEQUENCE / LIMIT

- Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks
- parent material does not alone determine every soil property.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Weathering is controlled not by climate alone but by the interaction of energy and moisture with mineralogy, discontinuities, relief, biota, time and land use.

02

STAGE 02

SOIL EROSION SEQUENCE AND WATERSHED CONTROL LOGIC

SOIL EROSION SEQUENCE AND WATERSHED CONTROL LOGIC · RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION DIFFUSE · CHAMBAL CUE · WATERSHED RESPONSE COVER SOIL · SLOW RUNOFF · SHORTEN SLOPE LENGTH · PROTECT DRAINAGE · TREAT GULLY HEAD

RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION diffuse sheet removal small rill channels entrenched → WATERSHED RESPONSE cover soil → slow runoff → shorten slope length → contour/terrace → protect drainage → treat gully head → rehabilitate catchment → WIND BRANCH surface creep → saltation → suspension → COASTAL BRANCH wave erosion belongs to the coastal-process system; use only when asked.

CORE CLAIM

- RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION diffuse sheet removal small rill channels entrenched gullies ravine landscape: Chambal cue
- WATERSHED RESPONSE cover soil → slow runoff → shorten slope length → contour/terrace

MECHANISM

- protect drainage → treat gully head → rehabilitate catchment
- WIND BRANCH surface creep → saltation → suspension

CONSEQUENCE / LIMIT

- COASTAL BRANCH wave erosion belongs to the coastal-process system; use only when asked.
- A check dam alone cannot replace upstream cover, drainage and maintenance.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Rainwater harvesting becomes effective only when capture, water quality, aquifer suitability, maintenance and demand control are treated as one urban water-budget system.

03

STAGE 03

MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY MECHANISM

MASS-MOVEMENT TAXONOMY AND SLOPE-STABILITY MECHANISM · MASS MOVEMENT! · SLOW SLIDE/FALL FLOW/AVALANCHE CREEP, SOLIFLUCTION ROCKFALL DEBRIS FLOW TRANSLATIONAL · SLOPE BALANCE RESISTING STRENGTH · COHESION + FRICTION + STRUCTURAL SUPPORT VERSUS DRIVING STRESS · GRAVITY COMPONENT + LOADING + TOE LOSS RAIN INFILTRATION · ADDED WEIGHT + PORE PRESSURE · LOWER EFFECTIVE STRENGTH ROAD OR RIVER CUT

MASS MOVEMENT

SLOW SLIDE/FALL FLOW/AVALANCHE creep, solifluction rockfall debris flow translational slide → earth or mud flow rotational slump rock/debris avalanche snow avalanche → SLOPE BALANCE resisting strength = cohesion + friction + structural support versus driving stress =

CORE CLAIM

- MASS MOVEMENT
- SLOW SLIDE/FALL FLOW/AVALANCHE creep, solifluction rockfall debris flow translational slide
- earth or mud flow rotational slump rock/debris avalanche snow avalanche

MECHANISM

- SLOPE BALANCE resisting strength = cohesion + friction + structural support versus driving stress =
- gravity component + loading + toe loss rain infiltration → added weight + pore pressure
- → lower effective strength road or river cut → steepening / undercutting → support loss

CONSEQUENCE / LIMIT

- quake / quarry / leakage / drainage change → trigger or amplification
- FAILURE WHEN DRIVE RESISTANCE
- Material, movement and water define the event; speed alone does not.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Landslides occur at a threshold where driving stress exceeds resisting strength; water is especially important because it can simultaneously add weight, raise pore pressure and weaken material.

04

STAGE 04

INDIA LANDSLIDE GEOGRAPHY, RISK PRODUCTION AND MITIGATION STACK

INDIA LANDSLIDE GEOGRAPHY, RISK PRODUCTION AND MITIGATION STACK · INDIA LANDSLIDE GEOGRAPHY · YOUNG RELIEF + FRACTURES + INCISION + RAIN + · WESTERN GHATS · ESCARPMENT + DEEP WEATHERING + OROGRAPHIC RAIN + CUTS · WEATHERED SLOPES + ROADS + SETTLEMENTS + DRAINAGE CHANGE · GSI LANDSLIDE INVENTORY · GSI LADDER INVENTORY · GOSS GEOLOGY/SLOPE RAIN/QUAKE SLIDE/FLOW ROAD/TOWN DISASTER

INDIA LANDSLIDE GEOGRAPHY

Himalaya / NE : young relief + fractures + incision + rain + seismicity → Western Ghats : escarpment + deep weathering + orographic rain + cuts → Nilgiris : weathered slopes + roads + settlements + drainage change

CORE CLAIM

- INDIA LANDSLIDE GEOGRAPHY
- Himalaya / NE : young relief + fractures + incision + rain + seismicity
- Western Ghats : escarpment + deep weathering + orographic rain + cuts
- Nilgiris : weathered slopes + roads + settlements + drainage change
- PREDISPOSITION → TRIGGER → MOVEMENT → EXPOSURE → LOSS geology/slope rain/quake slide/flow road/town disaster

MECHANISM

- GSI LADDER inventory → susceptibility/hazard study → post-event diagnosis
- monitoring / NLFC support → decision communication
- MITIGATION STACK avoid unsafe siting → manage drainage → protect toe
- stabilise slope → bioengineer → control spoil/quarrying
- maintain works → warn/close routes → evacuate and prepare

CONSEQUENCE / LIMIT

- Inventory maps past events; susceptibility maps relative likelihood
- neither is a deterministic event prediction.
- Joshimath is a multi-causal subsidence case, not a synonym for landslide.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India needs a layered landslide strategy—avoid unsafe exposure, control water, stabilise diagnosed slopes, maintain infrastructure, and connect forecasts to last-mile action.

05

STAGE 05

GROUNDWATER ARCHITECTURE FROM RECHARGE TO WELLS

GROUNDWATER ARCHITECTURE FROM RECHARGE TO WELLS · SOIL MOISTURE · GROUND SURFACE RECHARGE AREA SPRING AT SLOPE · ZONE OF AERATION PERCHED WATER ON LOCAL AQUITARD CAPILLARY · WATER TABLE UNCONFINED AQUIFER PUMPING WELL · ----------> CONE OF DEPRESSION · AQUITARD SLOW TRANSMISSION · CONFINED AQUIFER PRESSURE; WATER RISES TO POTENTIOMETRIC LEVEL

rain → infiltration → soil moisture → percolation → recharge

GROUND SURFACE recharge area spring at slope → ZONE OF AERATION perched water on local aquitard capillary fringe → WATER TABLE UNCONFINED AQUIFER pumping well

CORE CLAIM

- rain → infiltration → soil moisture → percolation → recharge
- GROUND SURFACE recharge area spring at slope
- ZONE OF AERATION perched water on local aquitard capillary fringe
- WATER TABLE UNCONFINED AQUIFER pumping well

MECHANISM

- ----------> cone of depression
- AQUITARD slow transmission
- CONFINED AQUIFER pressure; water rises to potentiometric level
- AQUIC/LIDE practically no transmission porosity = fraction of voids

CONSEQUENCE / LIMIT

- ? permeability = connected flow capacity aquifer stores and transmits
- aquitard transmits slowly artesian water rises under pressure: it flows only if head exceeds ground
- Budget: change in storage = recharge + inflow - discharge - pumping - outflow.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Groundwater is a dynamic stock governed by recharge, discharge, pumping, inter-aquifer flow and quality; annual recharge is therefore not a licence to abstract the same quantity everywhere.

06

STAGE 06

INDIA GROUNDWATER DEPLETION, QUALITY AND SUBSIDENCE GEOGRAPHY

INDIA GROUNDWATER DEPLETION, QUALITY AND SUBSIDENCE GEOGRAPHY · AQUIFER FORM · MAJOR PRESSURE · LAYERED ALLUVIUM · DEPLETION, AS, NITRATE · WEATHERED FRACTURES · DISCONTINUOUS STORAGE · DECCAN BASALT

REGION	AQUIFER FORM	MAJOR PRESSURE
Indo-Gangetic	layered alluvium	depletion, As, nitrate
Peninsular	weathered fractures	discontinuous storage
Deccan basalt	contacts + fractures	uneven well yield
Arid hard rock	limited recharge	fluoride, salinity
Coastal belt	fresh-saline wedge	intrusion and upconing crop-water demand + cheap energy + private wells + urban growth weak aquifer regulation + limited recharge falling head → lower yield → higher lift cost → unequal access compressible sediment: pore pressure loss → compaction → subsidence

CORE CLAIM

- AQUIFER FORM
- MAJOR PRESSURE
- Arid hard rock
- layered alluvium
- depletion, As, nitrate
- weathered fractures

MECHANISM

- contacts + fractures
- uneven well yield
- Arid hard rock
- limited recharge
- fluoride, salinity

CONSEQUENCE / LIMIT

- intrusion and upconing crop-water demand + cheap energy + private wells + urban growth
- weak aquifer regulation + limited recharge falling head → lower yield → higher lift
- cost → unequal access compressible sediment: pore-pressure loss → compaction → subsidence
- Quality paths differ: arsenic redox
- fluoride rock-water interaction nitrate sewage/fertiliser

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Good geography answers move from process to spatial pattern, then from consequence to a mechanism-matched and institutionally realistic response.

07

STAGE 07

RECHARGE, DEMAND GOVERNANCE AND INTEGRATED MAINS ANSWER SPINE

RECHARGE, DEMAND GOVERNANCE AND INTEGRATED MAINS ANSWER SPINE · GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS · RECHARGE DEMAND · GOVERNANCE ROOF AND RUNOFF CAPTURE CROP-WATER ALIGNMENT MANAGED AQUIFER · AQUIFER PLAN · CGWB ASSESSMENT + NAQUIM MAPPING + STATE REGULATION · ATAL BHUJAL LEARNING + LOCAL INSTITUTIONS + PUBLIC DATA · ANSWER SPINE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS • SUPPLY / RECHARGE DEMAND / GOVERNANCE roof and runoff capture crop-water alignment managed • aquifer recharge efficient irrigation watershed treatment well registration/metering spring-shed restoration conjunctive use and • reuse quality-safe source water community water budgets maintenance and monitoring enforceable state/local rules • AQUIFER PLAN	• Atal Bhujal learning + local institutions + public data • ANSWER SPINE • 1 define stock-flow-quality system • 2 draw aquifer or India map • 3 diagnose regional mechanism	• ? match intervention to cause • 6 quality limits and equity • Recharge structures are inputs, not proof of water-table recovery. • Desalination supplies water but does not restore freshwater aquifer head.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Desalination supplies water but does not restore freshwater aquifer head.

08

STAGE 08

EVIDENCE LADDER AND CONFIDENCE CONTROL

EVIDENCE LADDER AND CONFIDENCE CONTROL · EVIDENCE LADDER · FROM SOURCE BOOK · STANDARD KNOWLEDGE · CURRENT AFFAIRS · 2. AGRICULTURE.MD USES "CHERNOZEM", "ALLUVIAL" AND "BLACK SOIL" AS · 3. EVIDENCE UNITS AVAILABLE IN THIS FILE · SOURCE TYPE CONTROLS THE STRENGTH AND LIMITS OF THE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
EVIDENCE LADDER 1. = from source book + inference / standard knowledge + current affairs.	2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as... 3. Evidence units available in this file	VERDICT: Source type controls the strength and limits of the historical claim. EVIDENCE LADDER

CORE CLAIM

- EVIDENCE LADDER
- 1. = from source book + inference / standard knowledge + current affairs.

MECHANISM

- 2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as...
- 3. Evidence units available in this file

CONSEQUENCE / LIMIT

- VERDICT: Source type controls the strength and limits of the historical claim.
- EVIDENCE LADDER

MECHANISM / VERDICT — VERDICT: Source type controls the strength and limits of the historical claim.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Source type controls the strength and limits of the historical claim.

09

STAGE 09

EXAMINER TRAPS AND CONTESTED BOUNDARIES

EXAMINER TRAPS AND CONTESTED BOUNDARIES · CLOSE DISTINCTIONS · 1. CAUSAL METHOD INITIAL CONDITION · ENERGY/FORCE OR DRIVER → · 2. SCALE, INTERACTING CONTROL OR EVIDENCE LIMIT · 3. UPSC TRAP · A LANDSLIDE CAN DELIVER SEDIMENT TO A RIVER · QUALIFICATION EARNS MARKS; FALSE CERTAINTY IS A HARD FAILURE.

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CLOSE DISTINCTIONS 1. Causal method Initial condition → energy/force or driver →... 2. SCALE, INTERACTING CONTROL OR EVIDENCE LIMIT	1. UPSC TRAP / ANSWER-USE: A landslide can deliver sediment to a river,... VERDICT: Qualification earns marks; false certainty is a hard failure. CLOSE DISTINCTION: Weathering differs from erosion, porosity from permeability, water table from piezometric surface.	aquifer from aquitard, and slump/slide/flow/creep require movement-specific geometry rather than the generic label landslide.

CORE CLAIM

- CLOSE DISTINCTIONS
- 1. Causal method Initial condition → energy/force or driver →...
- 2. SCALE, INTERACTING CONTROL OR EVIDENCE LIMIT

MECHANISM

- 3. UPSC TRAP / ANSWER-USE: A landslide can deliver sediment to a river,...
- VERDICT: Qualification earns marks; false certainty is a hard failure.
- CLOSE DISTINCTION: Weathering differs from erosion, porosity from permeability, water table from piezometric surface.

CONSEQUENCE / LIMIT

- aquifer from aquitard, and slump/slide/flow/creep require movement-specific geometry rather than the generic label landslide.

MECHANISM / VERDICT — VERDICT: Qualification earns marks; false certainty is a hard failure.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Qualification earns marks; false certainty is a hard failure.

10

STAGE 10

INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION

INTEGRATED ANSWER SPINE AND QUALIFIED CONCLUSION · ANSWER SPINE · 1. ORIGINAL SOLVED MAINS QUESTIONS 6 · 2. LOCAL OFFICIAL PAPERS/KEYS CHECKED · 2018, 2019, 2020, 2021, 2023 · 3. PRACTICE CONTRACT EVERY SOLVED ITEM HAS DEMAND DECODING · CLAIM, NAMED EVIDENCE, ANALYSIS AND QUALIFICATION LEAD TO A · EVIDENCE LIMIT

ANSWER SPINE

1. Original solved Mains questions 6 → 2. Local official papers/keys checked: 2018, 2019, 2020, 2021, 2023 → 3. Practice contract Every solved item has demand decoding, a detailed

CORE CLAIM

- ANSWER SPINE
- 1. Original solved Mains questions 6
- 2. Local official papers/keys checked: 2018, 2019, 2020, 2021, 2023,...

MECHANISM

- 3. Practice contract Every solved item has demand decoding, a detailed f...
- VERDICT: Claim, named evidence, analysis and qualification lead to a graded...
- EVIDENCE LIMIT: A trigger such as intense rain is not the complete cause: antecedent moisture, discontinuities, toe cutting,

CONSEQUENCE / LIMIT

- ? groundwater trends and notified areas require dated official status.

MECHANISM / VERDICT — VERDICT: Claim, named evidence, analysis and qualification lead to a graded...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Claim, named evidence, analysis and qualification lead to a graded.

11

STAGE 11

EVIDENCE LADDER AND CONFIDENCE CONTROL — DEEP-REVIEW SYNTHESIS 2

EVIDENCE LADDER AND CONFIDENCE CONTROL — DEEP-REVIEW SYNTHESIS 2 · EVIDENCE LADDER · FROM SOURCE BOOK · STANDARD KNOWLEDGE · CURRENT AFFAIRS · 2. AGRICULTURE.MD USES "CHERNOZEM", "ALLUVIAL" AND "BLACK SOIL" AS · 3. EVIDENCE UNITS AVAILABLE IN THIS FILE · SOURCE TYPE CONTROLS THE STRENGTH AND LIMITS OF THE

EVIDENCE LADDER → 1. = from source book + inference / standard knowledge + current affairs. → 2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as → 3. Evidence units available in this file → VERDICT: Source type controls the strength and limits of the historical claim.

CORE CLAIM

- EVIDENCE LADDER
- 1. = from source book + inference / standard knowledge + current affairs.

MECHANISM

- 2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as...
- 3. Evidence units available in this file

CONSEQUENCE / LIMIT

- VERDICT: Source type controls the strength and limits of the historical claim.
- EVIDENCE LADDER

MECHANISM / VERDICT — VERDICT: Source type controls the strength and limits of the historical claim.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Pedogenesis begins with weathered parent material but becomes soil formation only when organic activity, translocation and time create an organised profile; black soil illustrates how basaltic parent material and seasonal moisture regimes shape properties.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND · ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK · ADVANCED 3 — FORECAST SKILL, FALSE ALARMS AND NON-STATIONARITY · ADVANCED 4 — AQUIFER HETEROGENEITY, CAPTURE AND SUSTAINABLE YIELD · ADVANCED 5 — GOVERNANCE EVALUATION · IN TECHNICAL TERMS, SHEAR STRENGTH IS OFTEN REPRESENTED BY · COHESION PLUS EFFECTIVE NORMAL STRESS MULTIPLIED BY FRICTION.

OPTIONAL SOURCE DEPTH

- ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND THRESHOLDS
- ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK
- ADVANCED 3 — FORECAST SKILL, FALSE ALARMS AND NON-STATIONARITY
- ADVANCED 4 — AQUIFER HETEROGENEITY, CAPTURE AND SUSTAINABLE YIELD

ADVANCED DEBATE

- ADVANCED 5 — GOVERNANCE EVALUATION
- ADVANCED 1 — EFFECTIVE STRESS, FACTOR OF SAFETY AND THRESHOLDS The simple core rule is driving stress versus resisting strength.
- In technical terms, shear strength is often represented by a Mohr–Coulomb relation: cohesion plus effective normal stress multiplied by friction.
- Pore pressure lowers effective normal stress.

LEGACY / NUANCE

- A factor of safety above one indicates calculated resistance exceeds driving force; near one means marginal stability.
- Real slopes remain uncertain because geometry, groundwater pathways and material properties vary.
- ADVANCED 2 — WEATHERING PROFILES, CLAYS AND HYDRO-MECHANICAL FEEDBACK Weathering
- fronts can create saprolite, corestones, clay-rich horizons and permeability contrasts.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.